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1.

$$\alpha = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$$

$$\Rightarrow \alpha = \sqrt{2 + \alpha}$$

$$\Rightarrow \alpha^2 - \alpha - 2 = 0$$

$$\Rightarrow (\alpha - 2)(\alpha + 1) = 0 \quad \therefore \alpha = 2 \quad (\alpha > 0)$$

2.

$$a_{n+1} = 3 - \frac{1}{a_n} \quad n \rightarrow \infty \text{ 이면 } \alpha = 3 - \frac{1}{\alpha} \text{ 이다.}$$

$$\text{즉, } \alpha^2 - 3\alpha + 1 = 0 \text{ 이다. } \therefore \alpha = \frac{3 \pm \sqrt{5}}{2}$$

$$\text{여기서 } a_1 = 1, a_2 = 2, a_3 = \frac{5}{2} \dots \text{ 이므로 } \alpha = \frac{3 + \sqrt{5}}{2} \text{ 이다.}$$

3.

유계인 함수 $f(x)$ 가 주어진 구간에서 유한개의 불연속점을 갖으면
상적분 하적분이 일치하므로 적분이 가능하다.

⑧ $f(x) = [x], I = [0, 3]$ (단, $[x]$ 는 x 보다 크지 않은 가장 큰 정수)는
 $x = 1, 2, 3$ 에서 불연속이다. 즉, 유한개의 불연속점을 갖는다.
 $f(x) = [x]$ 는 구간 $I = [0, 3]$ 에서 적분가능하다.

⑨ $f(x) = \begin{cases} 1 & (0 < x < 1) \\ 0 & (x = 0, 1) \end{cases}, I = [0, 1]$ 은 $x = 0, 1$ 에서
2개의 불연속점을 갖는다. 즉, 유한개의 불연속점을 갖는다.
그러므로 함수 $f(x)$ 는 적분가능하다.

⑩ $f(x) = \begin{cases} 1 & (x \text{는 유리수}) \\ 0 & (x \text{는 무리수}) \end{cases},$
 $I = [0, 1]$ 는 무한개의 불연속점이 생기므로 적분이 불가능하다.

4.

$$f(x) = \begin{cases} x^2 & x \in [0, 1] \cap Q \\ -x^2 & x \in [0, 1] \cap Q^c \end{cases} \text{ 의 상적분은 } \int_0^1 x^2 dx = \frac{1}{3} \text{ 이다.}$$

5.

$$\Delta x = \frac{\pi}{3} \text{ 이므로 } f(x) = \ln(2 + \cos x) \text{ 에서}$$

$$x_0 = 0 \Rightarrow f(x_0) = \ln 3$$

$$x_1 = \frac{\pi}{3} \Rightarrow f(x_1) = \ln\left(2 + \frac{1}{2}\right) = \ln 5 - \ln 2$$

$$x_2 = \frac{2}{3}\pi \Rightarrow f(x_2) = \ln\left(2 - \frac{1}{2}\right) = \ln 3 - \ln 2$$

$$x_3 = \pi \Rightarrow f(x_3) = \ln(2 - 1) = 0$$

$$x_4 = \frac{4}{3}\pi \Rightarrow f(x_4) = \ln\left(2 - \frac{1}{2}\right) = \ln 3 - \ln 2$$

$$x_5 = \frac{5}{3}\pi \Rightarrow f(x_5) = \ln\left(2 + \frac{1}{2}\right) = \ln 5 - \ln 2$$

$$x_6 = 2\pi \Rightarrow f(x_6) = \ln(2 + 1) = \ln 3 \text{ 이므로}$$

$$\int_0^{2\pi} \ln(2 + \ln x) dx$$

$$\approx \frac{\pi}{9} \{ \ln 3 + 2(\ln 5 - \ln 2) + 4(\ln 3 - \ln 2) + 2 \cdot 0 \\ + 4(\ln 3 - \ln 2) + 2(\ln 5 - \ln 2) + \ln 3 \} = \frac{2}{9} \pi$$

6.

$$y = \sqrt[3]{1-x^5} \text{ 이라 할 때,}$$

$$x = \sqrt[3]{1-y^5} \Leftrightarrow x^3 = 1-y^5 \Leftrightarrow y^5 = 1-x^3 \Leftrightarrow y = \sqrt[5]{1-x^3} \text{ 이므로}$$

$$y = \sqrt[3]{1-x^5} \text{ 와 } y = \sqrt[5]{1-x^3} \text{ 은 역함수 관계이다.}$$

$$\int_a^b f(x) dx + \int_{f(a)}^{f(b)} f^{-1}(x) dx = bf(b) - af(a) \text{ 이므로}$$

$$\int_0^1 (\sqrt[3]{1-x^5} - \sqrt[5]{1-x^3} + 1) dx$$

$$= \int_0^1 \sqrt[3]{1-x^5} dx - \int_0^1 \sqrt[5]{1-x^3} dx + \int_0^1 1 dx$$

$$= \int_0^1 \sqrt[3]{1-x^5} dx + \int_1^0 \sqrt[5]{1-x^3} dx + \int_0^1 1 dx$$

$$= 0 - 0 + \int_0^1 1 dx = 1 \text{ 이다.}$$

7.

$$\lim_{n \rightarrow \infty} \left(\frac{8n}{n^2 + 0^2} + \frac{8n}{n^2 + 1^2} + \frac{8n}{n^2 + 2^2} + \dots + \frac{8n}{n^2 + (n-1)^2} \right)$$

$$= 8 \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{n}{n^2 + k^2}$$

$$= 8 \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{\frac{1}{n}}{1 + \left(\frac{k}{n}\right)^2} = 8 \int_0^1 \frac{1}{1+x^2} dx$$

$$= 8 [\tan^{-1} x]_0^1 = 8 \left(\frac{\pi}{4} - 0 \right) = 2\pi$$

8.

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n k^2 (\ln \sqrt[n]{(n+k)^3} - \ln \sqrt[n]{n^3})$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n k^2 \left(\frac{3}{n} \ln(n+k) - \frac{3}{n} \ln n \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k}{n} \right)^2 \frac{3}{n} \cdot \ln \left(1 + \frac{k}{n} \right)$$

$$= \int_0^1 3x^2 \ln(1+x) dx$$

$$= [x^3 \ln(1+x)]_0^1 - \int_0^1 x^3 \frac{1}{1+x} dx \quad (\because \text{부분적분})$$

$$= \ln 2 - \int_0^1 \frac{x^3 + 1}{1+x} - \frac{1}{1+x} dx$$

$$= \ln 2 - \int_0^1 x^2 - x + 1 - \frac{1}{1+x} dx$$

$$= \ln 2 - \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 + x - \ln(1+x) \right]_0^1$$

$$= 3 \left(\frac{2}{3} \ln 2 - \frac{5}{18} \right) = 2 \ln 2 - \frac{5}{6}$$

9.

$$\lim_{n \rightarrow \infty} \frac{\left(\sum_{k=1}^n k^3 \right) \left(\sum_{k=1}^n k^5 \right)}{\left(\sum_{k=1}^n k \right) \left(\sum_{k=1}^n k^7 \right)} = \lim_{n \rightarrow \infty} \frac{\left(\sum_{k=1}^n \left(\frac{k}{n} \right)^3 \frac{1}{n} \right) \left(\sum_{k=1}^n \left(\frac{k}{n} \right)^5 \frac{1}{n} \right)}{\left(\sum_{k=1}^n \left(\frac{k}{n} \right) \frac{1}{n} \right) \left(\sum_{k=1}^n \left(\frac{k}{n} \right)^7 \frac{1}{n} \right)}$$

$$= \frac{\int_0^1 x^3 dx \int_0^1 x^5 dx}{\int_0^1 x dx \int_0^1 x^7 dx} = \frac{\frac{1}{4} \cdot \frac{1}{6}}{\frac{1}{2} \cdot \frac{1}{8}} = \frac{2}{3} \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right)$$

[다른 풀이]

$$\lim_{n \rightarrow \infty} \frac{\left(\sum_{k=1}^n k^3 \right) \left(\sum_{k=1}^n k^5 \right)}{\left(\sum_{k=1}^n k \right) \left(\sum_{k=1}^n k^7 \right)} = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{4} n^4 \right) \left(\frac{1}{6} n^6 \right)}{\left(\frac{1}{2} n^2 \right) \left(\frac{1}{8} n^8 \right)} = \frac{\frac{1}{4} \cdot \frac{1}{6}}{\frac{1}{2} \cdot \frac{1}{8}} = \frac{2}{3}$$

10.

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \left\{ \left(\sum_{k=1}^n \sqrt{1 + \cos \frac{\pi k}{2n}} \right) \sqrt{1 - \cos \frac{\pi}{2n}} \right\} \\
 &= \lim_{n \rightarrow \infty} \left\{ \left(\sum_{k=1}^n \sqrt{2} \sqrt{\frac{1 + \cos \left(\frac{\pi k}{2n} \right)}{2}} \right) \sqrt{2} \sqrt{\frac{1 - \cos \left(\frac{\pi}{2n} \right)}{2}} \right\} \\
 &= 2 \lim_{n \rightarrow \infty} \left\{ \left(\sum_{k=1}^n \sqrt{\frac{1 + \cos \left(\frac{\pi k}{2n} \right)}{2}} \right) \sqrt{\frac{1 - \cos \left(\frac{\pi}{2n} \right)}{2}} \right\} \\
 &= 2 \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n \cos \left(\frac{\pi k}{4n} \right) \sin \left(\frac{\pi}{4n} \right) \right\} \\
 &= 2 \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n \cos \left(\frac{\pi k}{4n} \right) \frac{1}{n} n \sin \left(\frac{\pi}{4n} \right) \right\} \\
 &= 2 \lim_{n \rightarrow \infty} \sum_{k=1}^n \cos \left(\frac{\pi k}{4n} \right) \frac{1}{n} \lim_{n \rightarrow \infty} n \sin \left(\frac{\pi}{4n} \right) \\
 &= 2 \int_0^1 \cos \left(\frac{\pi}{4} x \right) dx \lim_{n \rightarrow \infty} \frac{\sin \left(\frac{\pi}{4n} \right)}{\frac{\pi}{4n} \cdot \frac{4}{\pi}} \\
 &= \frac{\pi}{2} \int_0^1 \cos \left(\frac{\pi}{4} x \right) dx \\
 &= \frac{\pi}{2} \cdot \frac{4}{\pi} \left[\sin \left(\frac{\pi}{4} x \right) \right]_0^1 \\
 &= \sqrt{2}
 \end{aligned}$$

11.

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \left(\frac{n^2}{n^2 + 1^2} \right)^{\frac{1}{n^2 + 1^2}} \left(\frac{n^2}{n^2 + 2^2} \right)^{\frac{2}{n^2 + 2^2}} \cdots \left(\frac{n^2}{n^2 + n^2} \right)^{\frac{n}{n^2 + n^2}} \\
 &= \lim_{n \rightarrow \infty} e^{\ln \left(\frac{n^2}{n^2 + 1^2} \right)^{\frac{1}{n^2 + 1^2}} \left(\frac{n^2}{n^2 + 2^2} \right)^{\frac{2}{n^2 + 2^2}} \cdots \left(\frac{n^2}{n^2 + n^2} \right)^{\frac{n}{n^2 + n^2}}} \\
 &= e^{-\frac{1}{4} (\ln 2)^2} \\
 &(\because) \\
 & \lim_{n \rightarrow \infty} \ln \left(\frac{n^2}{n^2 + 1^2} \right)^{\frac{1}{n^2 + 1^2}} \left(\frac{n^2}{n^2 + 2^2} \right)^{\frac{2}{n^2 + 2^2}} \cdots \left(\frac{n^2}{n^2 + n^2} \right)^{\frac{n}{n^2 + n^2}} \\
 &= \lim_{n \rightarrow \infty} \ln \left(\frac{n^2}{n^2 + 1^2} \right)^{\frac{1}{n^2 + 1^2}} + \ln \left(\frac{n^2}{n^2 + 2^2} \right)^{\frac{2}{n^2 + 2^2}} + \cdots + \ln \left(\frac{n^2}{n^2 + n^2} \right)^{\frac{n}{n^2 + n^2}} \\
 &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{n^2}{n^2 + k^2} \right)^{\frac{k}{n^2 + k^2}} \\
 &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n^2 + k^2} \ln \left(\frac{n^2}{n^2 + k^2} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{1 + \left(\frac{k}{n} \right)^2} \left(\frac{k}{n} \right) \frac{1}{n} \ln \left(\frac{1}{1 + \left(\frac{k}{n} \right)^2} \right) \\
 &= \int_0^1 \frac{x}{1 + x^2} \ln \left(\frac{1}{1 + x^2} \right) dx = - \int_0^1 \frac{x}{1 + x^2} \ln (x^2 + 1) dx \\
 &= - \frac{1}{4} \left[\{ \ln (x^2 + 1) \}^2 \right]_0^1 = - \frac{1}{4} (\ln 2)^2
 \end{aligned}$$

12.

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \sum_{k=n+1}^{2n} \frac{3\sqrt{k}}{n\sqrt{n}} \\
 &= 3 \int_1^2 x^{\frac{1}{2}} dx \quad \left(\because \frac{k}{n} = x, \frac{1}{n} = dx \right) \\
 &= 2 \left[x^{\frac{3}{2}} \right]_1^2 \\
 &= 2(2\sqrt{2} - 1) \\
 &= 4\sqrt{2} - 2
 \end{aligned}$$

13.

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \cos^3 \left(\frac{x}{2} \right) \sin^3 \left(\frac{x}{2} \right) dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{8} \left(2 \cos \left(\frac{x}{2} \right) \sin \left(\frac{x}{2} \right) \right)^3 dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{8} \sin^3 x dx \quad (\because \text{2배각 공식}) \\
 &= \frac{1}{8} \cdot \frac{2}{3} \times 1 = \frac{1}{12} \quad (\because \text{왈리스 정리})
 \end{aligned}$$

14.

$$\begin{aligned}
 & \frac{\int_0^1 (1 - x^2)^{2020} dx}{\int_0^1 (1 - x^2)^{2019} dx} \\
 &= \frac{\int_0^{\frac{\pi}{2}} \cos^{4040} \theta \cos \theta d\theta}{\int_0^{\frac{\pi}{2}} \cos^{4038} \theta \cos \theta d\theta} \quad (\because x = \sin \theta) \\
 &= \frac{\int_0^{\frac{\pi}{2}} \cos^{4041} \theta d\theta}{\int_0^{\frac{\pi}{2}} \cos^{4039} \theta d\theta} = \frac{\frac{4040}{4041} \cdot \frac{4038}{4039} \cdots \frac{2}{3} \cdot 1}{\frac{4038}{4039} \cdot \frac{4036}{4037} \cdots \frac{2}{3} \cdot 1} = \frac{4040}{4041}
 \end{aligned}$$

15.

$$\begin{aligned}
 & \int_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} \cos^{-1} x dx = [x \cos^{-1} x]_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} - \int_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} x \frac{-1}{\sqrt{1-x^2}} dx \quad (\because \text{기함수}) \\
 &= \frac{1}{\sqrt{5}} \cos^{-1} \left(\frac{1}{\sqrt{5}} \right) - \left(-\frac{1}{\sqrt{5}} \right) \cos^{-1} \left(-\frac{1}{\sqrt{5}} \right) \\
 &= \frac{1}{\sqrt{5}} \left(\cos^{-1} \left(\frac{1}{\sqrt{5}} \right) + \cos^{-1} \left(-\frac{1}{\sqrt{5}} \right) \right) \\
 &= \frac{\pi}{\sqrt{5}}
 \end{aligned}$$

[다른 풀이]

$$\begin{aligned}
 & \int_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} \cos^{-1} x dx = \int_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} \frac{\pi}{2} - \sin^{-1} x dx = \int_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} \frac{\pi}{2} dx = \frac{\pi}{\sqrt{5}} \\
 & \text{평균값} = \frac{\int_{-\frac{1}{\sqrt{5}}}^{\frac{1}{\sqrt{5}}} \cos^{-1} x dx}{\frac{1}{\sqrt{5}} - \left(-\frac{1}{\sqrt{5}} \right)} = \frac{\frac{\pi}{\sqrt{5}}}{\frac{2}{\sqrt{5}}} = \frac{\pi}{2}
 \end{aligned}$$

16.

$$\begin{aligned}
 & y' = \sqrt{1-x} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2} \sqrt{\frac{1}{x} - 1} \text{ 이므로} \\
 & x = \frac{1}{2} \text{ 에서 접선의 기울기는 } y' \left(\frac{1}{2} \right) = \frac{1}{2} \text{ 이다.} \\
 & x = \frac{1}{2} \text{ 일 때,} \\
 & y = \int_0^{\frac{1}{\sqrt{2}}} \sqrt{1-t^2} dt \\
 &= \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta \quad (t = \sin \theta \text{ 치환}) \\
 &= \int_0^{\frac{\pi}{4}} \frac{1 + \cos 2\theta}{2} d\theta = \left[\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{\frac{\pi}{4}} = \frac{\pi + 2}{8} \\
 & \text{이므로 접선의 방정식은 다음과 같다.} \\
 & y = \frac{1}{2} \left(x - \frac{1}{2} \right) + \frac{\pi + 2}{8} = \frac{1}{2} x + \frac{\pi}{8}
 \end{aligned}$$

17.

$$\begin{aligned} A'(t) &= \frac{1}{2}(\sinh^2 t + \cosh^2 t) - \sqrt{\cosh^2 t - 1} \sinh t \\ &= \frac{1}{2}(\sinh^2 t + \cosh^2 t) - \sinh t \\ &= \frac{1}{2}(\cosh^2 t - \sinh^2 t) = \frac{1}{2} \end{aligned}$$

18.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1}{x} \int_x^{2x} g(t) dt \\ &= \lim_{x \rightarrow 0} \frac{\int_x^{2x} g(t) dt}{x} \quad \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{g(2x) \times 2 - g(x)}{1} \\ &= 2g(0) - g(0) \\ &= g(0) = (f^{-1})'(0) \\ &= \frac{1}{f'(1)} = \frac{1}{1+e} \end{aligned}$$

19.

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\int_0^h \sin(t^2) dt}{h} \\ &= \lim_{h \rightarrow 0} \int_0^h \sin(t^2) \frac{dt}{h^3} \\ &= \lim_{h \rightarrow 0} \frac{\sin(h^2)}{3h^2} \quad (\because \text{로피탈 정리}) \\ &= \frac{1}{3} \end{aligned}$$

20.

라이프니츠 정리에 의해

$$\begin{aligned} f'(x) &= \sin(x+x^4) \cdot 2x - \sin(x+1) \cdot 0 + \int_1^{x^2} \cos(x+t^2) dt \text{이다.} \\ \therefore f'(1) &= \sin 2 \cdot 2 - \sin 2 \cdot 0 + 0 = 2\sin 2 \end{aligned}$$

21.

$$\begin{aligned} \int_0^x (x^2 - t^2) f(t) dt &= x^4 \ln \left(\frac{x^4}{e} \right) \text{에서 양변을 } x \text{로 미분하면} \\ \Rightarrow \int_0^x 2xf(t) dt &= 4x^3 \ln \left(\frac{x^4}{e} \right) + 4x^3 \\ \Rightarrow \int_0^x f(t) dt &= 2x^2 \left(\ln \left(\frac{x^4}{e} \right) + 1 \right) \\ \Rightarrow \int_0^x f(t) dt &= 8x^2 \ln x \text{ 양변을 } x \text{로 미분하면} \\ \Rightarrow f(x) &= 16x \ln x + 8x, \therefore f(1) = 8 \end{aligned}$$

22.

1) $0 \leq x < 1$

$$F(x) = \int_0^x 2t dt = x^2$$

2) $1 \leq x \leq 3$

$$\begin{aligned} F(x) &= \int_0^1 2t dt + \int_1^x f(t) dt \\ &= \int_0^1 2t dt + \int_1^x (2-t) dt \\ &= 1 + \left[2t - \frac{1}{2}t^2 \right]_1^x \\ &= 1 + 2x - \frac{1}{2}x^2 - 2 + \frac{1}{2} \\ &= -\frac{1}{2}x^2 + 2x - \frac{1}{2} \end{aligned}$$

$$\text{따라서 } F(x) = \begin{cases} x^2, & 0 \leq x < 1 \\ -\frac{1}{2}x^2 + 2x - \frac{1}{2}, & 1 \leq x \leq 3 \end{cases} \text{이다.}$$

① $\lim_{x \rightarrow 1} F(x) = 1 = F(1)$ 이므로 $F(x)$ 구간 $[0, 3]$ 에서 연속이다.

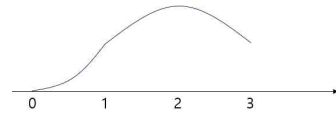
② $F'(x) = \begin{cases} 2x, & 0 \leq x < 1 \\ -x + 2, & 1 \leq x \leq 3 \end{cases}$ 이고,

$$\lim_{h \rightarrow 1-} \frac{F(1+h) - F(1)}{h} = 2$$

$$\lim_{h \rightarrow 1+} \frac{F(1+h) - F(1)}{h} = 1 \text{ 이므로 } x=1 \text{에서 미분불가능하다.}$$

③ 구간 $[1, 3]$ 에서 $F(x)$ 는 증가하다가 감소한다.

④ $F(x)$ 의 그래프 개형을 그려보면 $x=2$ 에서 최댓값을 갖는다.



23.

$$f(x) = \int_0^{x^2} |\cos t| dt \text{ 일 때, } f'(x) = 2x |\cos x^2| \text{ 이고}$$

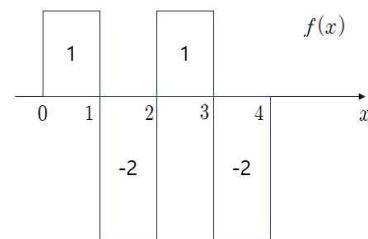
$$f'(x) = \begin{cases} \cos(x^2) \cdot 2x, & 0 < x \leq \sqrt{\frac{\pi}{2}} \\ -\cos(x^2) \cdot 2x, & \sqrt{\frac{\pi}{2}} < x < \sqrt{\frac{3\pi}{2}} \end{cases} \text{이므로}$$

$$f''(x) = \begin{cases} -\sin(x^2) \cdot 4x^2 + 2\cos(x^2), & 0 < x < \sqrt{\frac{\pi}{2}} \\ \sin(x^2) \cdot 4x^2 - 2\cos(x^2), & \sqrt{\frac{\pi}{2}} < x < \sqrt{\frac{3\pi}{2}} \end{cases}$$

$$\begin{aligned} &= \begin{cases} g(x), & 0 < x < \sqrt{\frac{\pi}{2}} \\ h(x), & \sqrt{\frac{\pi}{2}} < x < \sqrt{\frac{3\pi}{2}} \end{cases} \\ g\left(\sqrt{\frac{\pi}{2}}\right) &\neq h\left(\sqrt{\frac{\pi}{2}}\right) \text{이므로 } f''\left(\sqrt{\frac{\pi}{2}}\right) \text{은 존재하지 않는다.} \end{aligned}$$

24.

$$f(x) = \begin{cases} 1 & (0 \leq x < 1) \\ -2 & (1 \leq x < 2) \end{cases} \text{ 이고 } g(x) = \int_0^x f(t) dt \text{ 이므로}$$



$$\sum_{n=1}^{10} g\left(n + \frac{1}{2}\right) = g\left(\frac{3}{2}\right) + g\left(\frac{5}{2}\right) + g\left(\frac{7}{2}\right) + \dots + g\left(\frac{21}{2}\right)$$

$$= 0 + \left(-\frac{1}{2}\right) + (-1) + \left(-\frac{3}{2}\right) + \dots + \left(-\frac{9}{2}\right)$$

$$= -\frac{1}{2}(1 + 2 + 3 + \dots + 9)$$

$$= -\frac{1}{2} \frac{9(10)}{2} = -\frac{45}{2} \text{이다.}$$

25.

$$f(t) = \begin{cases} -4\left[\frac{t}{\pi}\right] + 2 & (0 \leq t \leq 2\pi) \\ -4\left[\frac{t-2\pi}{\pi}\right] + 2 & (2\pi \leq t \leq 4\pi) \end{cases}$$

$$= \begin{cases} 2 & (0 \leq t < \pi) \\ -2 & (\pi \leq t < 2\pi) \\ 2 & (2\pi \leq t < 3\pi) \\ -2 & (3\pi \leq t < 4\pi) \end{cases} \text{이고}$$

$$g(x) = \begin{cases} \frac{1}{\pi} \int_0^x \left(-4 \left\lfloor \frac{t}{\pi} \right\rfloor + 2 \right) dt & (0 \leq t < 2\pi) \\ \frac{1}{\pi} \int_0^x \left(-4 \left\lfloor \frac{t-2\pi}{\pi} \right\rfloor + 2 \right) dt & (2\pi \leq t < 4\pi) \end{cases} \quad \text{이므로}$$

$0 \leq t < \pi$ 일 때,

$$g(x) = \frac{1}{\pi} \int_0^x f(t) dt = \frac{1}{\pi} \int_0^x 2 dt = \frac{2}{\pi} x$$

$\pi \leq t < 2\pi$ 일 때,

$$\begin{aligned} g(x) &= \frac{1}{\pi} \int_0^x f(t) dt \\ &= \frac{1}{\pi} \int_0^{\pi} 2 dt + \frac{1}{\pi} \int_{\pi}^x (-2) dt \\ &= -\frac{2}{\pi} x + 4 \end{aligned}$$

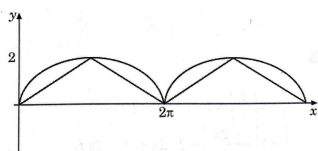
$2\pi \leq t < 3\pi$ 일 때,

$$\begin{aligned} g(x) &= \frac{1}{\pi} \int_0^x f(t) dt \\ &= \frac{1}{\pi} \int_0^{\pi} 2 dt + \frac{1}{\pi} \int_{\pi}^{2\pi} (-2) dt + \frac{1}{\pi} \int_{2\pi}^x 2 dt \\ &= \frac{2}{\pi} x - 4 \end{aligned}$$

$3\pi \leq t < 4\pi$ 일 때,

$$\begin{aligned} g(x) &= \frac{1}{\pi} \int_0^x f(t) dt \\ &= \frac{1}{\pi} \int_0^{\pi} 2 dt + \frac{1}{\pi} \int_{\pi}^{2\pi} (-2) dt + \frac{1}{\pi} \int_{2\pi}^{3\pi} 2 dt + \frac{1}{\pi} \int_{3\pi}^x (-2) dt \\ &= -\frac{2}{\pi} x + 8 \end{aligned}$$

따라서 $0 \leq x \leq 4\pi$ 에서 $g(x)$ 의 그래프는 아래 그림의 두 삼각형이고, 곡선 C 는 삼각형에 외접하는 사이클로이드가 된다.



따라서 곡선 C 와 $g(x)$ 의 그래프 사이의 넓이는

$$2 \left(3\pi - \frac{1}{2} \times 2\pi \times 2 \right) = 2\pi$$

[참고] cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$

$(0 \leq \theta \leq 2\pi)$ 와 x 축 사이의 넓이는 $3\pi a^2$ 이다.